



Department of Computer Science and Business Systems

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: I Semester B.Tech/CSBS

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member: Mrs.B.Yazhini, AP/CSBS

Innovative Practice Description

- **Unit / Topic: Unit I / Illustrative Problem: Minimum in a list**
- **Course Outcome: CO1**
- **Topic Learning Outcome: TLO3**
- **Activity Chosen: Role play**
- **Justification:**

The role play helps the students to play and understand the basics of theories handled. The solving of Minimum in a list is an interesting game but the recursion concept behind this problem is difficult to understand. In order to provide the clear understanding of recursion the roleplay activity was conducted.
- **Time Allotted for the Activity: 10 Minutes**
- **Details of the Implementation:**
 - Role play is used to show a live demonstration of how a particular task will be handled. It gives a clear understanding to the students while they are involving in that activity.
 - The faculty member explained the concept of solving the minimum in a list.
 - 3 students are asked to play the role of source tower, destination tower and auxiliary tower. The game rules are explained to them. They have to follow the rules to solve the problem.
 - Student 1 (Mr.Nivas) act as number 8 in a list as shown in Figure 1. Based on the rule the other students 2 & 3 Mr. Asvathan and Mr.Naveen act as 4 and 6 respectively.
 - Figure 2 shows Mr.Parthiban who acts to find the minimum in list and how the number 3 is it could be explained by comparing his position with the prior student numbers.
 - From this the students clearly understood the concept of recursion clearly.
 - Figure 1 shows the students involvement in the activity.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO12
CO1	3	2	3	3	3	2	2	2

	(1 – Low	2 – Moderate	3 – High)
CO	PSO1	PSO2	PSO3
CO1	3	3	3

PSO3: PO / PSO mapped:

	PO1	PO2	PO3	PO 5	PO8	PO9
Innovative practice	3	2	3	3	3	2
Justification for correlation	The students will apply algorithmic approach to solve the various engineering problems by using basics mathematics , science and engineering principles	The algorithmic solutions can be identified, formulated to solve the simple and complex engineering problems.	The students will able to produce the potential design solutions using flowchart to meet the requirement and validate the design with different set of inputs.	The students will use flowgorithm software to develop pictorial representation of solution of the various real time problems.	The student will follow the rules of algorithm, pseudo code and flowchart for providing the solution to the problems.	The students will able to solve the various problem by using algorithmic design tools individually

	PO10	PO12	PSO1	PSO2	PSO3
Innovative practice	2	2	3	3	3
Justification for correlation	The students will able to explore their knowledge of solving the problems using algorithm, flowchart and pseudo code by giving presentation and through discussion forum	The students will continuously improve their knowledge in python programming concepts by completing online course..	The students will able to produce the potential design solutions using flowchart to meet the requirement and validate the design with different set of inputs.	The students will use flowgorithm software to develop pictorial representation of solution of the various real time problems.	The student will follow the rules of algorithm, pseudo code and flowchart for providing the solution to the problems.

- **Images / Screenshot of the practice:**



Fig1.1: The three students represent the three numbers in a list



Figure1.2 The Fourth student act to find the minimum in a list

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- They feel it is very interesting and told they understand how the to solve the Minimum in a list.
- The students are energetically involved and enjoyed the activity.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- It helps the students to understand the recursive procedure of solving the minimum in a list.
- This activity makes the class more interactive.

❖ ***Challenges faced in implementation:***

- All the students cannot be involved in the activity.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/CS8493-OS-Roleplay.pdf>
- https://www.ritrjpm.ac.in/images/computer-science/38_CS8391_Roleplay.pdf

Signature of Faculty Member

HOD/CSBS